MINI PROJECT
PHASE 1 REPORT

Name

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Project Title

Sales Transaction Analysis and Business Insights

Data Pre-processing and Preliminary Analysis using Excel

Tools Used: Microsoft Excel, Power Query and Pivot Tables

1. Introduction

This report explains the work completed in Phase 1 of the mini project. The main task in this phase was to take the raw Dirty Transaction Dataset, clean the data, make the values consistent, and prepare it for analysis. I used Excel Power Query for the cleaning and transformation work and then created Pivot Tables in Excel to get an initial understanding of the data.

2. Dataset Description

The dataset has 1,800 transaction records and 10 columns. It contains customer information, product category, country, transaction amount, date, and payment information. These columns give enough information to study sales and transaction patterns.

Column	Description
transaction_id	Unique ID for each transaction
customer_name	Name of the customer
email	Customer email
age	Age of the customer
country	Country
product_category	Product category
unit_price	Price of one unit
quantity	Number of units purchased
transaction_date	Date of the transaction
payment_method	Payment method used

3. Data Cleaning

The raw data was cleaned using Excel Power Query. I used Power Query because it made it easier to handle the data in a systematic way and apply the same transformations to the dataset.

3.1 Duplicate Check

The transaction data was checked for duplicate records. After checking the transaction information, the final cleaned data contained 1,800 records. Therefore, no transaction records were removed as duplicates.

3.2 Handling Missing Values

There were missing values in several columns. I handled them based on the type of data. For numerical columns, I used the median where a numerical replacement was appropriate. For categorical columns, I used values such as Unknown or Not Provided. For missing transaction dates, I did not create an artificial date and treated them as unknown.

3.3 Why Median Was Used

The median was used for age, unit price, and quantity because it is less affected by very high or very low values. The median values used were 43 for age, 1200 for unit price, and 2 for quantity.

4. Missing Value Details

Column	Missing Values	Action Taken
email	151	Replaced with Not Provided
age	141	Replaced with median value 43
unit_price	192	Replaced with median value 1200
quantity	180	Replaced with median value 2
transaction_date	83	Handled as Unknown/missing
payment_method	161	Replaced with Unknown

5. Data Standardization

The text fields were checked and standardized so that differences in capitalization, spacing, or formatting would not result in unnecessary categories. The data types were also checked so that numeric columns, text columns, and dates were stored in the correct format.

6. Creating the Total Sales Field

A calculated field called Total_Sales was created so that the total value of each transaction could be analyzed easily.

Total_Sales = unit_price × quantity

7. Fact and Dimension Tables

After cleaning the data, I split it into a fact table and dimension tables. The purpose was to keep the transaction values in the fact table and store descriptive information separately. This makes the data easier to manage and also prepares it for creating relationships in Power BI.

Table	Rows	Columns
Dim_Customer	1,766	customer_key, customer_name, email, age
Dim_Product	4	product_key, product_category
Dim_Country	6	country_key, country
Dim_Payment	4	payment_key, payment_method
Dim_Date	669	date_key, transaction_date, Year, Month, Month_Number, Quarter

7.1 Fact_Transactions Table

The final Fact_Transactions table contains 1,800 rows and 10 columns. The dimension keys were added to the fact table using Power Query Merge Queries so that each transaction can be connected to the correct dimension.

Column	Purpose
transaction_id	Transaction identifier
transaction_date	Transaction date
customer_key	Connects to Dim_Customer
product_key	Connects to Dim_Product
country_key	Connects to Dim_Country
payment_key	Connects to Dim_Payment
date_key	Connects to Dim_Date
unit_price	Numeric transaction value
quantity	Number of units
Total_Sales	Calculated transaction sales

7.2 Key Validation

The five key columns in the fact table were checked after the merge. There were no null values in customer_key, product_key, country_key, payment_key, or date_key.

8. Excel Pivot Tables

After completing the cleaning and table split, I created Pivot Tables to get a quick summary of the data. These Pivot Tables were useful for understanding the main sales patterns and deciding which information would be useful for the report.

Pivot Table	Rows	Values	Purpose
PT_Sales_Category	product_category	Total_Sales	Compare sales between categories
PT_Sales_Country	country	Total_Sales	Compare sales between countries
PT_Sales_Payment	payment_method	Total_Sales	Compare sales by payment method
PT_Monthly_Sales	Year / Month	Total_Sales	View sales over time

Pivot Table	Rows	Values	Purpose
PT_Quantity_Category	product_category	quantity	Compare quantity sold

9. Conclusion

In Phase 1, the raw transaction data was cleaned and organized using Excel Power Query. Missing values were handled, the data was standardized, Total_Sales was created, and the dataset was divided into fact and dimension tables. The Pivot Tables then gave an initial view of the sales data and completed the main Excel-based analysis work.